Margin

Overview

Live Demo

GitHub Repository

Time spent: Approximately 6.5 hours.

Stack: Next.js, TypeScript, Tailwind CSS, React-PDF / PDF.js.

The problem

I enjoy reading research, particularly about planetary science. Highlighting a paper isn't the same as working through it, and I've always strugguled with hving a **single source of truth** for my notes, questions, and highlights. I wanted to be able to capture a thought, ask a question, and return to the passage that prompted it. Margin is a reading workspace that keeps those things together, so the thinking I do while reading becomes something I can refer back to along side the origional souce material.

Key decisions

- **Keep the paper central.** The original PDF scrolls continuously and stays in place when panels change. The surrounding interface uses warm surfaces and restrained motion, while the PDF keeps its original appearance.
- **Attach thoughts to passages.** Selecting text starts a highlight, note, or question beside its source. Margin and list views offer different ways to review annotations; Hidden and Focus modes give the paper more room.
- **Make structure useful.** A compact section map combines navigation, reading position, and annotation markers. Expanding it reveals the contents, while previews help readers find relevant passages without losing their place.
- **Let the work leave the app.** Markdown, JSON, and CSV exports keep the quoted passage and its location available alongside the reader's notes.

Engineering choices

I used React-PDF rather than building a PDF renderer. Nearby pages mount their canvases and selectable text layers as needed, with page space reserved to keep scrolling stable. I paid attention to keyboard interactions, focus handling, and reduced-motion preferences. AI tools helped with layout exploration and implementation; I directed the product scope and refined the design and interactions.

Tradeoffs and assumptions

I prioritized the reading-and-annotation flow over document ingestion, accounts, and cloud sync. The demo uses a seeded library of 11 studies with sample annotations to demonstrate the interactions. New annotations and edits are session-only and reset on reload; reading position and display preferences persist locally. The reader assumes a desktop-sized screen and text-based PDFs. Smaller screens show a link to continue on a desktop.

With more time

I would first persist personal annotations and add PDF uploads, then develop richer understanding levels within highlights. I'd also explore Companion Mode and an optional chat integration with a bring-your-own-model approach, building on the reading workflow. Additionally, due to time, Annotaitons `List` view isn't as polished as I would have liked, but I decided to keep it in regardless.